

IILM UNIVERSITY

GREATER NOIDA, UTTAR PRADESH

CODEALPHA VIRTUAL INTERNSHIP

PYTHON PROGRAMMING

A 1-Month Professional Internship Presentation Report

INTERNSHIP DURATION

1 July 2026 – 30 July 2026 (1 Month)

STUDENT CREDENTIALS

Tejansh Vaibhav

Enrollment / Roll No.
25SCS1003005029

Program
B.Tech (CSE)

HOST ORGANIZATION

CodeAlpha
Virtual Python Programming Internship

Candidate & Internship Overview

Candidate Details

Student Name: TEJANSH VAIBHAV

Roll Number: 25SCS1003005029

Degree: B.Tech (Computer Science & Engineering)

Academic Batch: 2026–2027

University: IILM University, Greater Noida, U.P.

Internship Profile

Program Name: CodeAlpha Virtual Internship

Specialization: Python Programming

Duration: 1 Month (1 July – 30 July 2026)

Reference ID: CA/DF1/182298

Mode: Completely Online

Organization Profile: CodeAlpha

Practical Learning

CodeAlpha provided the Virtual Python Programming Internship with practical assignments and project-based learning.

Internship Format

The internship was completely online and focused on hands-on Python programming, implementation, testing and debugging.

Core Focus

Python fundamentals • Problem solving • Input validation • Data structures • File operations • Terminal-based applications • GitHub documentation

Problem Statement & Practical Context

The Challenge

Python concepts are often learned theoretically, but students need practical exposure to build complete user-interactive applications.

The Solution Scope

The internship addressed this gap through three working applications: Basic Chatbot, Stock Portfolio Tracker and Hangman Game.

Skills Applied

Input handling • Validation • Conditions • Loops • Functions • Data structures • File I/O • Testing • Debugging • Documentation

Core Project Objectives

- | | | |
|----|---------------------------------|--|
| 01 | Python Fundamentals | Strengthen syntax, functions, loops, conditions and modular programming. |
| 02 | Interactive Applications | Build practical terminal-based applications with clear user interaction. |
| 03 | Validation & Testing | Handle invalid input, edge cases and program flow reliably. |
| 04 | Data & Files | Use data structures and TXT/CSV file operations where required. |
| 05 | Problem Solving | Improve logical thinking, debugging and independent implementation. |
| 06 | Documentation | Maintain and present projects through GitHub repositories. |

Technologies & Tools Used

Python 3.x	Core programming language for all three projects.
Standard Library	random, csv, os and other built-in Python functionality.
Functions & Control Flow	Modular logic, loops, conditions and reusable functions.
Dictionaries & Data	Portfolio storage, lookups and structured application data.
File I/O	TXT and CSV export for the Stock Portfolio Tracker.
GitHub	Code hosting, repository organization and project documentation.

Python Project Architecture



Basic Chatbot

User Input → Validation → Response
Matching → Output → Continue /
Exit

Stock Portfolio Tracker

Stock Selection → Quantity →
Calculations → Summary → TXT / CSV
Export

Hangman Game

Random Word → Letter Input →
Validation → Game State → Win/Lose
→ Replay

Internship Timeline

Week 1

Planning & Fundamentals

Understood requirements, project flow and core Python concepts.

Week 2

Basic Chatbot

Implemented interaction, responses, validation and exit flow.

Week 3

Stock Portfolio Tracker

Implemented portfolio data, calculations, summaries and TXT/CSV export.

Week 4

Hangman & Testing

Implemented gameplay, validation, win/lose logic, replay and final testing.

Project Development Workflow

1

Requirement Understanding

Identify expected functionality and user flow for each project.

2

Implementation

Build modular Python functions and connect the application flow.

3

Validation & Testing

Test valid/invalid input, edge cases, game states and output formatting.

4

Documentation

Organize GitHub repositories and prepare final internship documentation.

Key Projects & Demonstrated Skills

Basic Chatbot

Continuous interaction, predefined responses, randomized greetings, input validation and modular functions.

Stock Portfolio Tracker

Stock selection, quantity validation, portfolio calculations, formatted summary and TXT/CSV export.

Hangman Game

Random word selection, letter validation, game-state tracking, win/lose detection, ASCII-art stages and replay.

Overall Skills

Python fundamentals, logical problem solving, debugging, testing, data handling, file I/O and GitHub documentation.

Certification & Official Verification

Official Record / Credential Details

Internship Title: Python Programming Virtual Internship

Issuing Organization: CodeAlpha

Mode: Completely Online

Reference / Student ID: CA/DF1/182298

Internship Period: 1 July 2026 – 30 July 2026

Student Roll No.: 25SCS1003005029

Original Internship Completion Certificate



Original Internship Offer Letter



INTERNSHIP OFFER LETTER

30th June 2026

STUDENT ID: CA/DF1/182298

Name: Tejansh Vaibhav

Welcome to CodeAlpha

Congratulations! We are delighted to make you an offer as **Python Programming Internship** offer letter is to confirm your selection **Python Programming Internship** at **CodeAlpha** and welcome you for the same effective from 1st July 2026 to 30th July 2026. Hope you will be doing well.

This Internship is observed by CodeAlpha as being a learning opportunity for you.

Your internship will embrace orientation and give emphasis on learning new skills with a deeper understanding of concepts through hands-on application of the knowledge you gained as an intern. You will acknowledge your obligation to perform all work allocated to you to the best of your ability within lawful and reasonable direction given to you.

We look forward to a worthwhile and fruitful association which will make you equipped for future projects.

Wishing you the most enjoyable and truly meaningful internship program experience.

Sincerely,


Swati Srivastava
Founder & CEO



Phone
+91 9336576683

Email
services@codealpha.tech

Address
Lucknow U.P

Original Letter of Recommendation



Website:
www.codealpha.tech

Email ID:
services@codealpha.tech
support@codealpha.tech

LinkedIn:
www.linkedin.com/company/codealpha

Mobile No:
+91 9336576683

CODEALPHA

YOUR DREAM OUR PASSION

Letter Of Recommendation

TO WHOM IT MAY CONCERN

This is to certify that Tejansh Vaibhav has successfully completed the Virtual Internship Program at CodeAlpha for a duration of one month from 1st July 2026 to 30th July 2026 as a Python Programming Internship.

He/She exhibited performance in this role and made a valuable contribution to our organization during the internship period. He/She has excellent analytical skills and He/She quickly acquired new skills and adeptly adapted to emerging technologies, demonstrating a high level of productivity.

He/She consistently displayed a willingness to provide assistance and fostered strong collaboration with fellow team members. His/Her dedication and skills make him/her a valuable asset to any prospective employer, and CodeAlpha wholeheartedly recommend him/her for any future endeavor.

REGARDS,

CA/DF1/182298

STUDENT I'D



FOUNDER & CEO



31st July 2026

DATE

Thank You!

Tejansh Vaibhav

Roll No: 25SCS1003005029

School of Computer Science & Engineering

IILM University, Greater Noida, U.P.